

GaussianBeV : 3D Gaussian Representation meets Perception Models for BeV Segmentation

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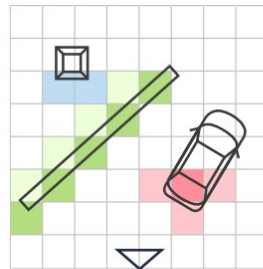
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- Problem/Objective

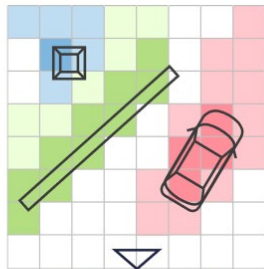
- BEV segmentation + GS

- Contribution/Key Idea

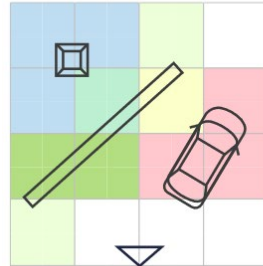
- *Optimization-free* image-to-3D gaussian representation 제안
- non scene-specific gaussian splatting



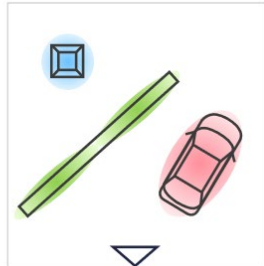
(a) Depth-based [8, 10, 13, 20]



(b) Projection-based [3, 7]

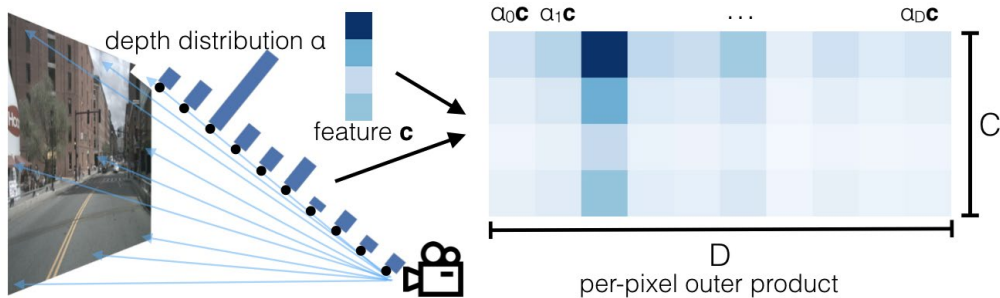


(c) Attention-based [17, 32]



(d) GaussianBeV (ours)

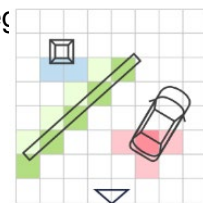
● Introduction - BEV



Depth-based method

● Depth-based method

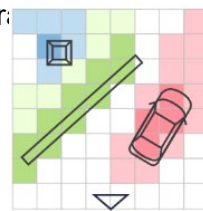
- ray sampling strategy



(a) Depth-based [8, 10, 13, 20]

● Projection-based method

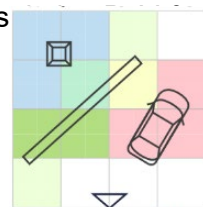
- all voxel along the ray share same feature



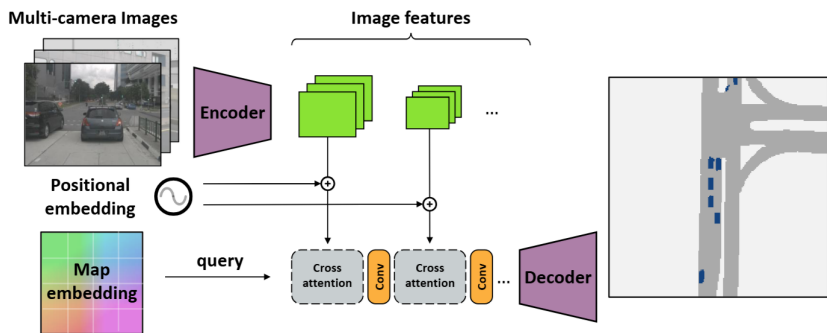
(b) Projection-based [3, 7]

● Attention-based method

- computational complexity & memory intensive
- 일부 low resolution psampling

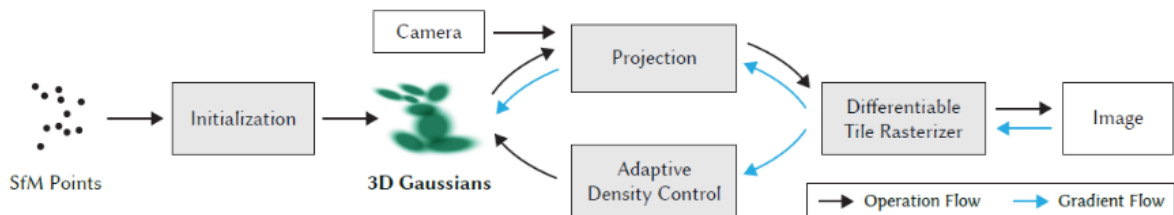


(c) Attention-based [17, 32]



Attention-based method

● Introduction - GS



- Positive)
 - (sparse voxel 에 비해) offer more efficient representation of scene
 - large gaussian으로 형상을 + small gaussian으로 fine detail을
- Drawback)
 - optimization requires many iterations → *optimization-free*
 - each scene마다 optimization → *learning NN capable of directly predicting a gaussian for any scene*

Method

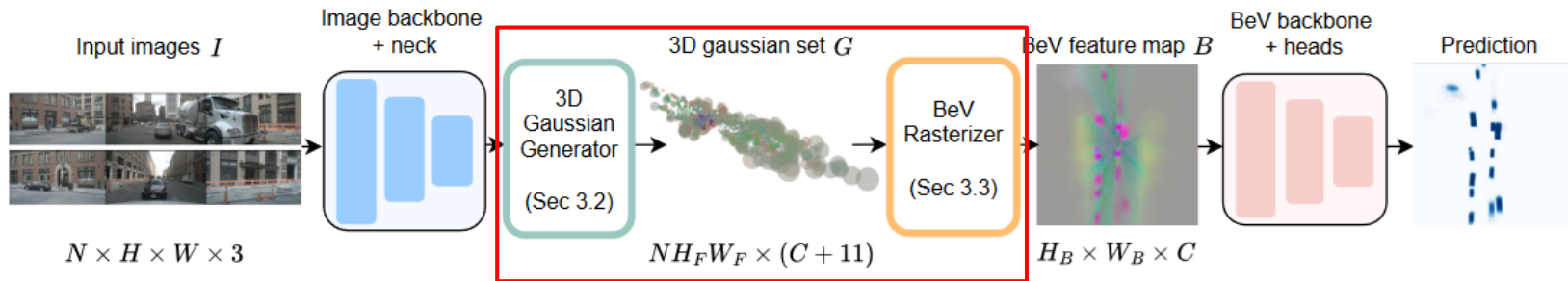


Figure 2. **Overview of GaussianBeV.** The network takes as input a set of multiview images and extracts features for each of them. The 3D gaussian generator module (Sec 3.2) predicts a 3D gaussian representation G of the scene which is then sent to the BeV rasterizer module (Sec 3.3) to performs BeV rendering. The resulting BeV feature map B is passed through a BeV backbone and segmentation heads to obtain the segmentation prediction. G and B are represented with colors only for visualization purpose.

- PV $\rightarrow$ BEV 네트워크를 GS 방식을 차용하여 대체

Method

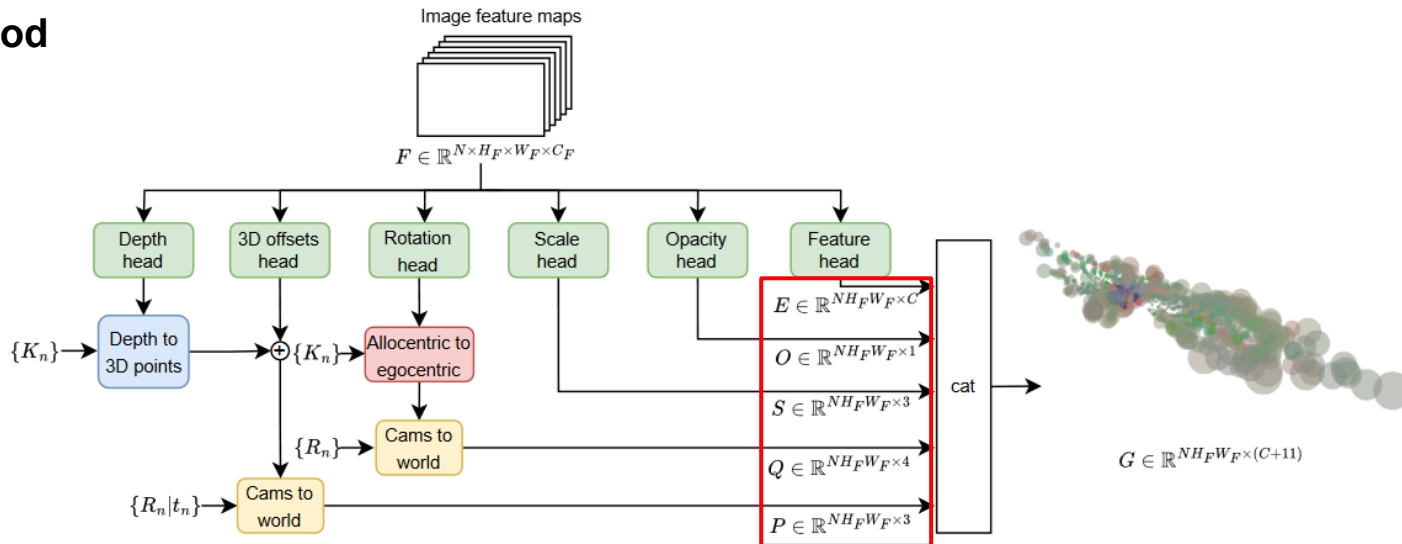


Figure 3. **3D Gaussian generator.** This module takes as input the feature map extracted from each camera, as well as the set of intrinsic $\{K_n\}$ and extrinsic $\{R_n|t_n\}$ parameters. For each pixel, it calculates the corresponding 3D gaussian by passing through prediction heads (green boxes). Some of these predictions are then decoded (blue and red boxes) and transformed to be expressed in the world reference frame (yellow boxes). All predicted gaussian parameters are then concatenated to produce the 3D gaussian representation G of the scene.

● Method - Position (3 - Dimension)

● Position : Depth + 3D offset

○ LiDAR Depth 를 사용하여 loss

$$z_{n,i} = \frac{f_n}{f} \left(\frac{1}{d_{n,i}} - 1 \right) \longrightarrow p_{n,i}^c = K_n^{-1} \begin{bmatrix} z_{n,i} u_{n,i} \\ z_{n,i} v_{n,i} \\ z_{n,i} \end{bmatrix}$$

→ 이렇게 하면 픽셀을 뚫어 나가는 ray (u,v,1) 위에만 존재

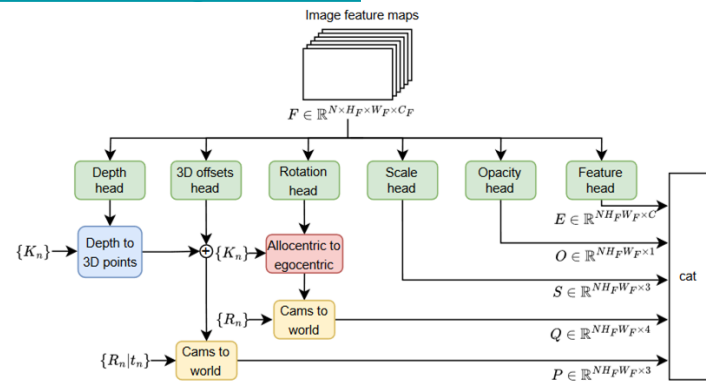
→ 근거리 3D offset을 학습하는 근거리 점으로
 $\Delta_{n,i} = (\Delta x_{n,i}, \Delta y_{n,i}, \Delta z_{n,i})^T$ 점으로

$$\bar{p}_{n,i}^c = p_{n,i}^c + \Delta_{n,i}$$

(카메라 좌표계)

$$p_{n,i}^w = [R_n | t_n] \bar{p}_{n,i}^c$$

카메라 → 월드 좌표계



$$P = \{p_{n,i}^w\} \in \mathbb{R}^{NH_F W_F \times 3}$$

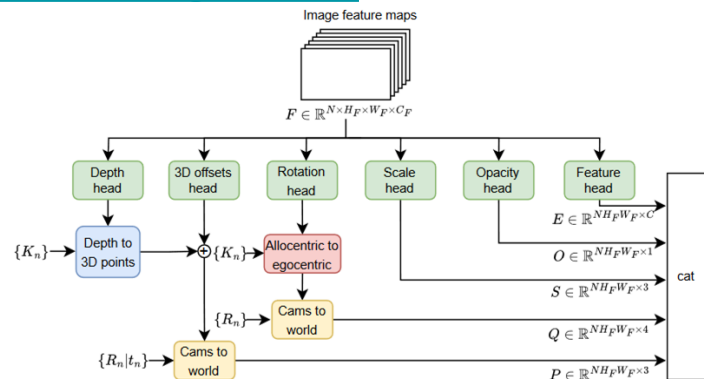
Method - Rotation (4 - Dimension)

- Rotation : allocentric rotation (3d optical ray)
 - z축 - ray 사이의 각도로 변환
 - "카메라 기준 z축" → "픽셀의 ray"로 가는 회전
- optical ray → Camera ego centric → World centric

$$q_{n,i}^e = q_{n,i} q_{n,i}^a$$



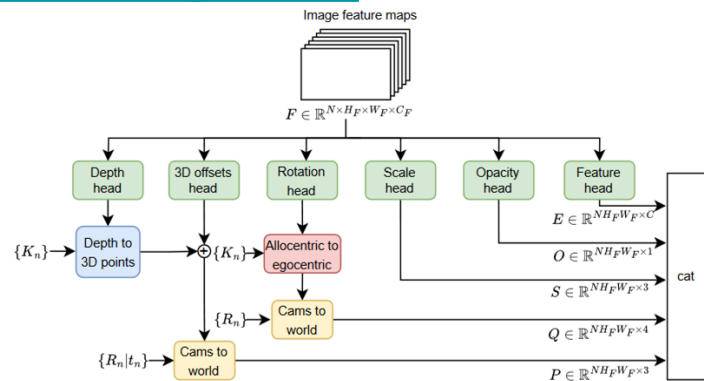
$$q_{n,i}^w = q_{R_n} q_{n,i}^e$$



$$Q = \{q_{n,i}^w\} \in \mathbb{R}^{NH_F W_F \times 4}$$

Method - Scale / Opacities / feature (3/1/C - Dimension)

- Scale
 - x, y, z 축에 대한 가우시안 분산
- Opacities
 - 가우시안 불투명도
- Feature
 - semantic feature embedding



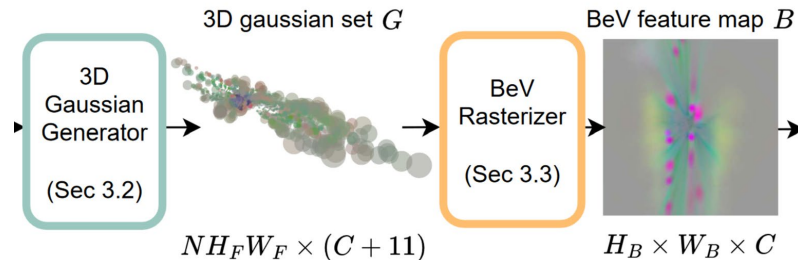
$$S \in \mathbb{R}^{NH_F W_F \times 3}$$

$$O \in \mathbb{R}^{NH_F W_F \times 1}$$

$$E \in \mathbb{R}^{NH_F W_F \times C}$$

Method - BEV Rasterizer

- Gaussian → BEV feature로의 변환 과정
 - 이때 합성 과정(=rasterization)은 미분 가능하게 설계
 - 즉, 학습/역전파 가능
- 기존 Gaussian Splatting은 색상을 렌더링
 - “semantic features”로 대체
 - 각 gaussian이 색상 x, perception에 중요한 high-dim feature vector
- 기존 perspective(원근)로 렌더링
 - BEV는 “탑다운” 뷰가 필요하므로 orthographic으로 변환



$$G \in \mathbb{R}^{NH_F W_F \times (C+11)}$$



$$B \in \mathbb{R}^{H_B \times W_B \times C}$$

- **Training**

$$L_{sem} = \lambda_{bce} L_{bce} + \lambda_{ctr} L_{ctr} + \lambda_{off} L_{off}$$

- Binary cross-entropy loss
- centerness loss
- offset loss

$$L = L_{sem} + L_{sem}^{early} + \lambda_{depth} L_{depth},$$

- Experiment

		all vehicles		vehicles with vis > 40%	
	Image backbone	224×448	448×800	224×448	448×800
FIERY [8]	EN-b4	35.8	-	39.8	-
CVT [32]	EN-b4	31.4	32.5	36.0	37.7
LaRa [1]	EN-b4	35.4	-	38.9	-
BEVFormer [15]	RN-50	35.8	39.0	42.0	45.5
Simple-BEV [7]	RN-50	36.9	40.9	43.0	46.6
BAEFormer [19]	EN-b4	36.0	37.8	38.9	41.0
PointBeV [3]	EN-b4	38.7	42.1	44.0	47.6
GaussianBeV	EN-b4	41.6	43.9	47.5	50.3

Table 1. GaussianBeV IoU results on vehicle segmentation on the nuScenes validation set using different input resolutions and different visibility filtering. Our method remarkably outperforms previous works on all experimental settings.

• Experiment

	Pedestrians
LSS [20]	15.0
FIERY [8]	17.2
ST-P3 [9]	14.5
TBP-Former static [5]	17.2
PointBEV [3]	18.5
GaussianBeV	21.2

Table 2. GaussianBeV IoU results on pedestrian segmentation on the nuScenes validation set. GaussianBeV is trained with 224×448 input image resolution and visibility filtering. Our method increases performances compared to previous works.

	Drivable area	Lane boundaries
LSS [20]	72.9	20.0
M ² BEV [26]	75.9	38.0
BEVFormer [15]	80.1	25.7
PETrv2 [17]	83.3	44.8
MatrixVT [33]	83.5	44.8
PointBeV [3] [†]	72.5	37.6
GaussianBeV	82.6	47.4

Table 3. GaussianBeV IoU results on both drivable area and lane boundary segmentation on the nuScenes validation set. GaussianBeV is trained with 448×800 input image resolution. [†] means that we trained the model by ourselves. Our method outperforms previous works on lane boundary segmentation and gives competitive results on drivable area segmentation.

• Experiment

Depth sup	Early sup	BeV backbone	rot	off	IoU
	✓	none	✓	✓	45.4
✓	✓	none	✓	✓	45.9
✓	✓	LSS [20]	✓	✓	47.0
		LSS [20]	✓	✓	46.9
		LSS [20]	✓	✓	46.7
✓	✓	LSS [20]			47.1
✓	✓	LSS [20]	✓		47.3
✓	✓	LSS [20]		✓	47.1
✓	✓	SimpleBeV [7]	✓	✓	47.0
✓	✓	LSS [20]	✓	✓	47.5

Table 4. Influence of BeV backbone, depth and early supervision, rotation and 3D offset prediction. Models are trained using 224×448 input resolution and visibility filtering on the vehicle segmentation task.

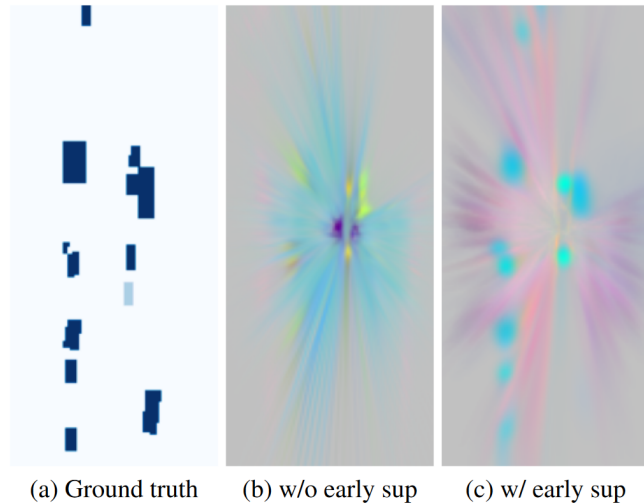


Figure 5. Influence of the early supervision. (a) Vehicle segmentation ground truth. (b) and (c) visualization of the BeV feature map for GaussianBeV trained without and with early supervision, respectively. Early supervision helps to achieve a representation closer to the content of the 3D scene. We observe that gaussians corresponding to the vehicles follow their shapes.

- Experiment

